

2G WIRELESS NETWORK

- ↳ 2G wireless N/w Marked the beginning of digital wireless communication.
- ↳ The switch to digital communication technology brought along other benefits such as high speed communication, data encryption, multiplexing capabilities, noise immunity, better call processing capabilities.

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GSM (GLOBAL SYSTEM FOR MOBILE)

- ↳ it is a 2G digital Cellular N/w.
- ↳ developed to solve the fragmentation problem of first generation Cellular System.
- ↳ it provides a common set of compatible services & capabilities to all mobile users worldwide.
- ↳ it was the world first Cellular System which specifies digital modulation & Network level architecture & services.

Basic Requirements

Service portability :- 1) The GSM must provide service portability.
2) Thus MS can access communication services in all participating countries.

Quality of Service :- The voice quality of GSM must be as good as previous Analog System.

Security :- The GSM system must provide secure on air transmission by information encryption without significantly increasing cost.

Radio frequency Utilization :- The GSM system must permit high spectrum efficiency & be capable of operating in the entire allocated freq. band.

NETWORK :- The GSM network plans must be based on relevant ITU recommendation.

Cost: The system parameter must be chosen with focus on reducing the cost of the complete GSM system.

GSM SERVICES

Bearer (Data) Service

- 1) These service related to layer 1, 2, 3 of OSI model.
- 2) provide data rates from 300bps to 9.6 Kbps
- 3) provide transparent, Non-transparent, ~~Bearer~~, ~~Non-Bearer~~ Synchronous, Non-Synchronous data transmission

Telephone Service

Include Voice oriented telephone service including encrypted voice transmission, message service & basic data comm.

also include Emergency Calling, facsimile, SMS, MMS soon.

Supplementary Services

like user identification, Call redirection, Call forwarding, multiparty comm.

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SYSTEM ARCHITECTURE

↳ It consist of three major interconnected subsystem

- 1) BSS (Base station subsystem)
- 2) NSS (Network & switching subsystem)
- 3) OSS (Operation support subsystem).

↳ 1) The BSS also known as the radio subsystem which support & manages radio transmission path b/w MS & MSC.

Mobile Station Mobile Switching Center

2) The BSS also manages the radio interference b/w the MS & other Subsystem of GSM System.

* Each BSS consist of many BSC (Base Station Controller) which are connected to MS to the NSS by MSC.

* ↳ The NSS Manages the switching function of the system & allows the MSC to communicate with other N/w such as PSTN & ISDN N/w

* ↳ The OSS supports the operation & maintenance of GSM & support System Engineer to monitor, diagnosis & troubleshoot all aspect of the GSM System.

This subsystem interacts with other GSM subsystem & is provided solely for the staff of the GSM operating system or company which provides service facilities for the N/w.

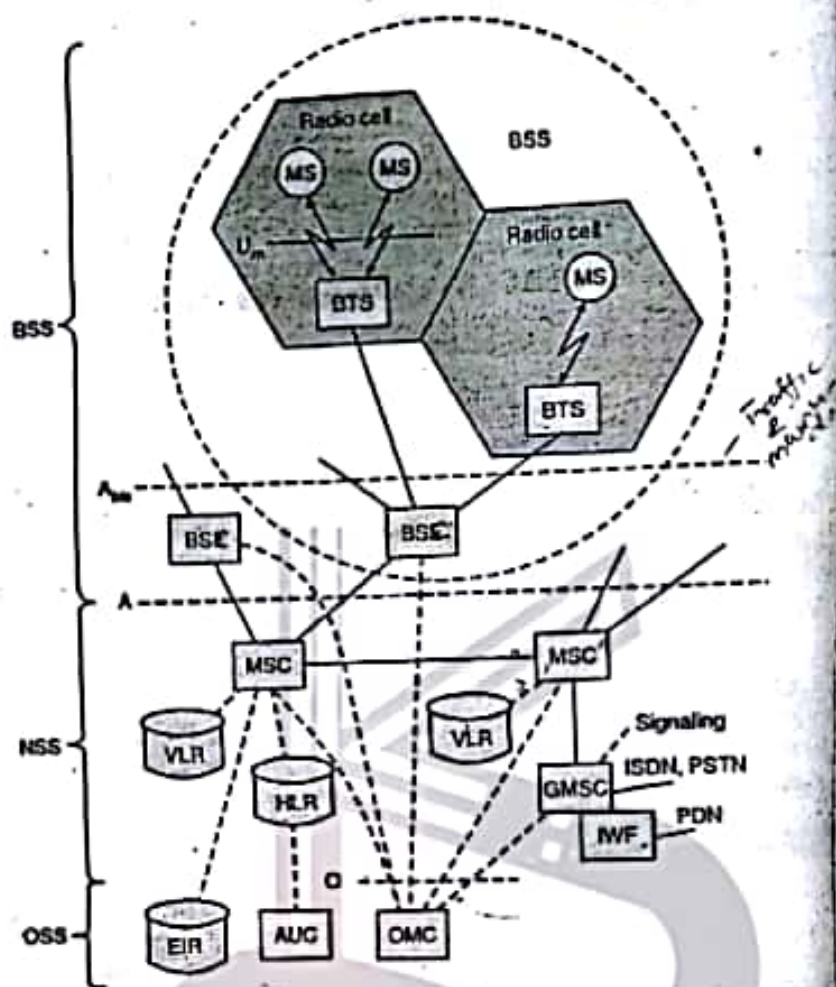


Fig. 5.2. GSM System Architecture.

Various Entities in GSM

1) BSS: Base station Subsystem. Perform all function necessary to maintain wireless communication connection to MS, Coding decoding of voice, rate adaption to wireless N/w apart.

1) BTS: Base transceiver station. Consist of all radio Equipment which are necessary for wireless transmission.
 it connects to MS via U_m interface
 to BSC via Abis interface

1.2) BSC :- Base Station Controller :-

1) The Control part in BSS is handled by BSC which manages many BTS.

2) Function

- ↳ radio frequency assignment
- ↳ handoff from one BTS to another.
- ↳ paging of the MS.

1.3) MS

2) NSS :- Network Subsystem

Various fn. such as Connection with standard Public N/w, handoff b/w diff. BSS's, support charging etc.

2.1) MSC

2.2) HLR

2.3) VLR

3) OSS :- Operation subsystem :-

perform various fn. for N/w operation & Maintenance

3.1) OMC (Operation & Maintenance Centre)

It monitors all other N/w Entities via O interface & perform fn. such as traffic monitoring

3.2) AUC (Authentication Centre)

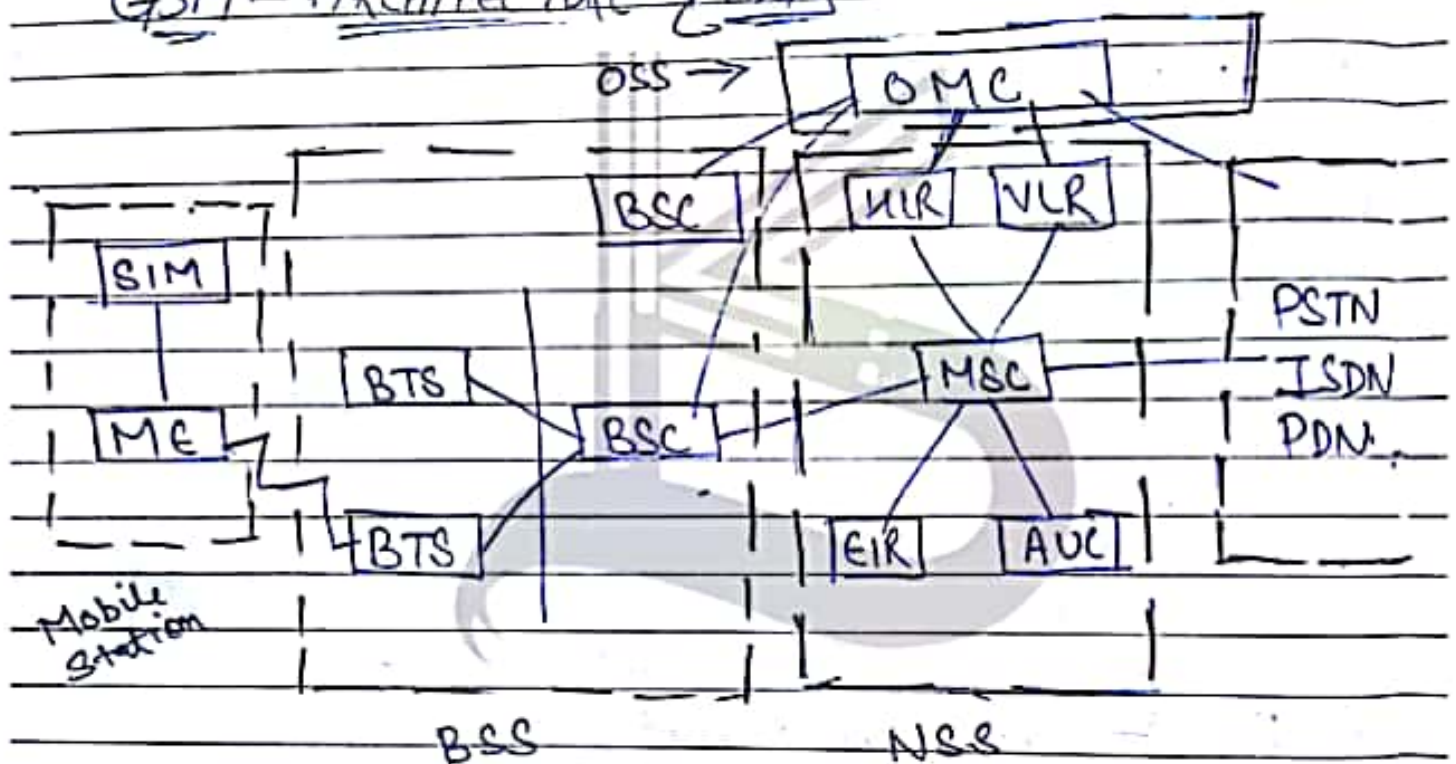
This system protects the user data identity & data transmission. It contains various Algo & Keys for authentication & encryption.

3.3) EIR (Equipment Identity Register)

This database store all the IMEIs of the devices registered for for the associated N/w.

for instance if MS is stolen, the ER block list the IMEI of that MS.

GSM - ARCHITECTURE DIAGRAM



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MOBILITY MANAGEMENT

Mobility Management Refers to the way the network manages the movement of the mobile subscribers which significantly affects the performance of the PS network.

HAND-OFF

Handoff is the process which enable a call to proceed uninterrupted when the user moves from one cell to another or one system to another.

There are two type of Handoff

↓
Intracell or inter-BS
Handoff

↓
InterSystem or inter-MSC
Handoff

INTER-BS HANDOFF

⇒ The Inter-BS Handoff Required when mobile unit travels from Coverage Region of one base station to Coverage Region of Another while new & old BS are connected to same MSC.

STEPS Followed:

- When the signal received from the current BS is not strong enough, the MS momentarily suspends conversation and initiates the handoff procedure by signaling on an idle channel in the new BS. Then, the conversation on the old BS is resumed.
- When MSC receives the signal, it transfers the information to the selected idle channel of the new BS and new conversation path is setup.
- After the MS has been transferred to the new BS, it signals the network and resumes conversation using the new channel.
- When MSC receives the handoff completion signal, the connection to old BS is terminated and the resources are released.

e) Mobile Call Origination

Steps

1) The MS sends the dialed no. to the MSC via BSC.

2) The MS Checks from VLR if MS is allowed the requested service.
If so; MSC asks BSC to allocate necessary resources.

3) If call is allowed MSC routes call to GMSC.

4) GMSC routes the call to local Exchange of called user.

5) The LE alerts (applies ringing) the called terminal.

6) Answer back (ring back tone) from called terminal to LE.

7) Answer back signal is routed back to MS through the serving MSC which also complete the speech path of MS.

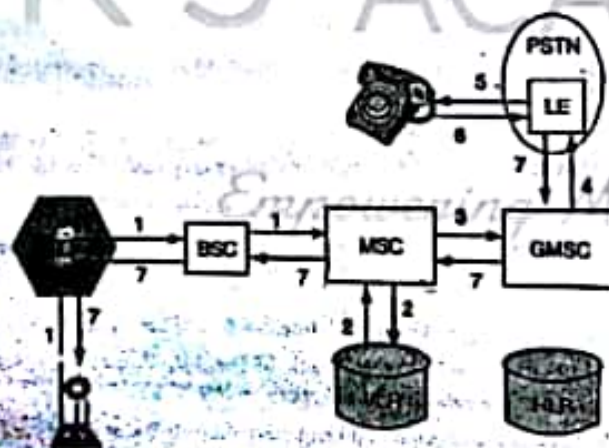


Fig. 5.10. Mobile Call Origination in GSM.

3) Mobile Call termination

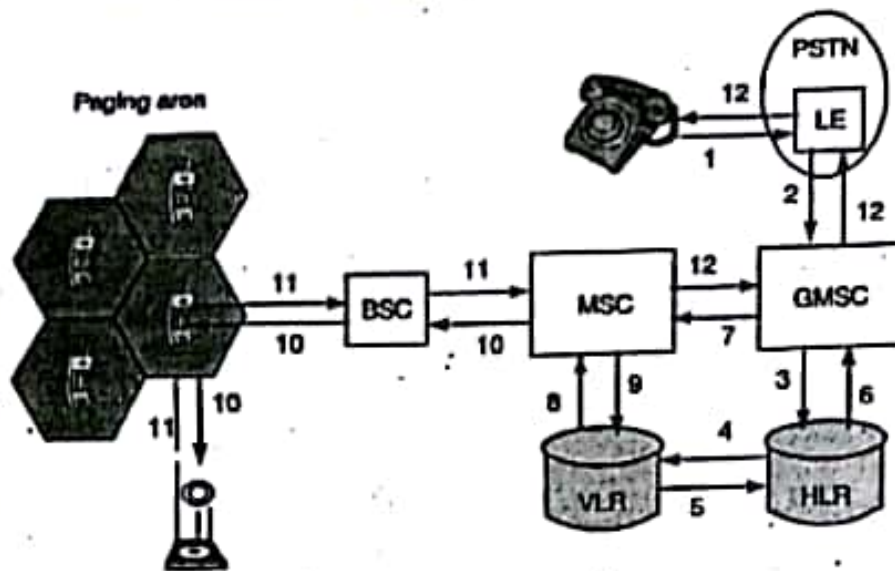


Fig. 5.11. Mobile Call Termination in GSM.

4) HANDOFF

In GSM, there are four possible handover scenarios:

- **Intra-cell handoff:** It occurs within a cell due to narrow-band interference. When this happens, the BSC changes the carrier frequency.
- **Inter-cell, intra-BSC handoff:** It occurs when the mobile station moves from one cell to another, but stays within the control of the same BSC.
- **Inter-BSC, intra-MSC handoff:** It happens when the mobile station leaves the coverage region of one BSC (which controls a limited number of cells) and enters the coverage region of another BSC. Here, the handoff has to be controlled by the MSC.
- **Inter MSC handoff:** It happens when the mobile station leaves the coverage region of one MSC and enters the coverage region of another MSC. Now, handoff process is controlled by both the MSCs together.

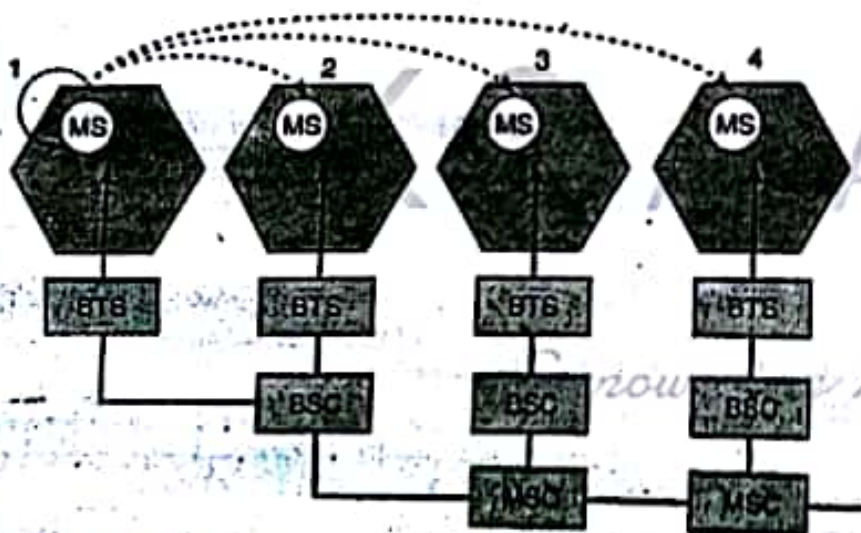


Fig. 5.12. Handoff Scenarios in GSM.